

COMP1682 – Final Year project

Final Project Report/Demo - Feedback sheet

Student Name and ID: Giulio Dajani and 001343717

Supervisor/Second Marker: Razia Sulthana Abdul Kareem

Overall Mark: 91

Overall Assessment:

1. Understanding of the Problem Domain

The project aims to develop a multimodal deepfake detection framework capable of distinguishing between real and fake videos under real-world conditions. The proposed system accepts both video and audio inputs, demonstrating a comprehensive approach toward multimodal analysis. The student has shown significant effort in understanding and analysing the problem domain, including the technical, ethical, and societal implications associated with deepfake detection. Ethical approval was formally obtained to collect and record necessary information related to the project, and appropriate consideration was given to legal, ethical, social, and professional issues relevant to the project. In addition, similar products and existing systems were comparatively studied, and a meaningful evaluation of related approaches was conducted to position the proposed work within the broader research landscape.

2. Development of Product Ideas and Implementation

The student conducted extensive analysis of suitable data collection techniques, tools, and evaluation methodologies throughout the development process. Strong evidence of complexity is observed in both the design and implementation stages of the project. The development process was highly iterative, involving multiple refinements and revisions to improve the detection of real and fake media content. The student maintained consistent engagement throughout the duration of the project and regularly discussed technical challenges, implementation issues, and testing outcomes during supervisory meetings. Several obstacles encountered during intermediate stages of development were systematically addressed and resolved successfully. The project demonstrates a clear incremental development approach, and the progression of the work has been consistently documented and reviewed.

3. Product Build and Evaluation

The student demonstrated strong technical capability in building the proposed system, and the project achieved the intended scope successfully. During the demonstration, the student responded confidently and accurately to detailed technical questions regarding the implementation, architecture, and codebase of the system. The project reflects substantial technical effort, particularly in integrating multimodal inputs and validating the effectiveness of the framework. An in-depth investigation and evaluation of the product were carried out during the demonstration, and the system was tested extensively with consideration for ethical and usability aspects. The student successfully demonstrated the usability, appropriateness, and practical applicability of the developed solution in addressing the targeted problem domain.

4. Conclusions and Critical Review

The student demonstrated critical thinking throughout the project and incorporated feedback effectively across multiple stages of report revision and system improvement. The report and project documentation were revised in response to constructive feedback, reflecting a strong commitment to continuous improvement and academic rigor. The student remained highly consistent, dedicated, and engaged throughout the lifecycle of the project, demonstrating professionalism and sustained effort in completing the work to a strong standard. At present, the system has been implemented primarily for laptop-based environments; however, the project could have been further extended by developing a minimal prototype or adaptive version suitable for mobile platform deployment.

Comments on the viva/demonstration

Have you held a demonstration or viva for this project? YES/NO

YES

Comments on the Standard of Presentation

Has the report met presentation criteria? YES/NO

YES

Mark Breakdown	
Each of the following four criteria carries equal weighting. Separate marks are awarded for both the report and demo/viva. The marks entered average to the overall grade awarded.	
1. Understanding of Problem Domain	
Report: 91	Demo\Viva: 91
Comments: Report: Demonstrates exceptional and authoritative understanding of the problem domain. The problem is precisely defined and convincingly justified within relevant academic, technical, and professional contexts. The literature review is comprehensive, current, and critically synthesised, clearly informing project aims and implementations. Methodology, requirements, tools, and development criteria are fully justified, with clear LSEPi consideration. The work demonstrates independent judgement, originality, and confident command of the domain, and is of publishable standard. Demo/Viva: Demonstrates a deep understanding of the domain problem, with strong awareness of existing work, key issues, and limitations. Project context, assumptions, constraints, and relevance are articulated with authority. Questions are answered with exceptional insight and critical perspective, drawing effectively on domain knowledge, literature, requirements, and comparable systems, demonstrating clear ownership of the problem and awareness of its wider implications.	
2. Development of Product and Ideas	
Report: 91	Demo\Viva: 91
Comments: Report: Demonstrates exceptional originality, autonomy and sophistication. The design process is systematic, iterative and strongly evidenced. Alternative approaches are critically evaluated, with rigorous justification for the selected solution. Data collection and analysis methods are appropriate, well executed and clearly inform design decisions. The work shows independent problem-solving, adaptability and a high level of product complexity that is clearly appropriate or ambitious for the project. The work is of a potentially publishable standard. Demo/Viva: Demonstrates exceptional understanding of the contextual development. Design choices, alternative approaches, trade-offs, and the evolution of the solution are articulated fluently and insightfully, reflecting professional maturity. Design and implementation decisions are clearly aligned with requirements and constraints.	
3. Product Build and Evaluation	
Report: 91	Demo\Viva: 91
Comments: Report: Demonstrates advanced technical competence with rigorous, well-justified evaluation. The product is substantial and potentially deployable, meeting or exceeding requirements. Implementation is clearly aligned with design and objectives, supported by systematic testing and critical evaluation using justified qualitative and quantitative metrics. Documentation offers clear insight into quality, limitations, and the significance of results. Overall, the work displays a high level of creativity, independence, originality, and strong potential for publication. Demo/Viva: Demonstrates exceptional technical proficiency and rigorous evaluation. Product is robust and very high standard. The work and findings are discussed with extensive objectivity and depth, showing insight and strong awareness of any limitations and areas for improvement. Responds rigorously to probing evaluation questions, shows the full scope of functionality with confidence and authority.	
4. Conclusions and Critical Review	
Report: 91	Demo\Viva: 91
Comments: Report: Demonstrates exceptional critical reflection and intellectual maturity. Conclusions are clearly derived from the evidence, implementation and evaluation, and are situated within the wider domain. Critically appraises strengths, limitations, trade-offs, mistakes and project decisions with honesty and depth. Lessons learned are specific and insightful. Future work is realistic, prioritised and justified, showing awareness of technical, methodological and professional implications. The reflection demonstrates autonomy, independent judgement and clear intellectual development. Demo/Viva: Demonstrates exceptional synthesis and critical reflection, evaluating successes, limitations, and future work with maturity. Conclusions are fully justified and grounded in evidence. Responses to questioning show clear insight into broader academic or professional implications and link conclusions directly to evidence and outcomes.	
Incremental Development	
Consistent	0

Clear and consistent evidence of incremental development of ideas, design, and product throughout the project. Progress is well-documented and verified through supervision meetings and shared repositories, showing continuous refinement and iteration.

COMP1682 – Final Year project

Final Project Report/Demo - Feedback sheet

Student Name and ID: Giulio Dajani (001343717)

Second Marker: Makuochi Nkwo

Broadly the final project should exhibit the following characteristics for the final grade. Both the report and the demonstration form a holistic project view and contribute to the criteria stated.

Marks	
≥ 90%	Meets all criteria. Shows a significant amount of critical analysis and exhibits an excellent understanding of the relevant issues. Product meets requirements.
80-89%	Meets almost all of the criteria. Demonstrates clear awareness and exposition of the relevant issues with a very high standard of critical analysis. Product meets requirements.
70-79%	Meets most of the criteria. Demonstrates clear awareness of relevant issues with a high standard of critical analysis. Appropriate use of analysis and design frameworks. Product meets all requirements.
60-69%	Meets most of the criteria. Demonstrates clear awareness of relevant issues with a very good standard of critical analysis. Very good analysis and design using appropriate frameworks but may include some errors. Product meets almost all of the requirements.
50-59%	Some of the criteria present, but mainly factual and descriptive, with some grasp of analysis. The analysis and design use appropriate frameworks but may have some errors. Product meets most of the requirements.
40-49%	The essential criteria present and establish a few relevant points, but superficial or confused exposition of issues with little grasp of analysis. An attempt has been made at analysis using appropriate frameworks, but there are many errors. Product meets the essential functional requirements.
<40	Little or no evidence of given criteria and no grasp of analysis. Does not demonstrate self-direction, originality in problem solving or a critical self-evaluation of project. Product meets very few of product requirements.

Overall Mark:

Overall assessment

The VerifAI system project aims to develop and evaluate a multimodal deepfake detection framework that improves cross-dataset generalisation, incorporates adaptive inference routing, and ensures reliable real-world deployment through systematic testing and API integration. Adopting a design science and experimental research methodology, the project employed adaptive multimodal detection that integrates feature-level fusion with dynamic inference routing to deliver a deployable, explainable system. The visual models include MobileNetV2, ViT, and Xception; the audio model uses ResNet18; and the fusion model employs an advanced deep multimodal architecture at the feature level. Libraries used include PyTorch (model development and training), OpenCV (extraction of video frames), Librosa (audio processing), FastAPI (backend deployment), and NumPy along with Pandas (data manipulation and analysis). This project is well scoped and solves a real problem. However, it consists of lots of moving components including video, audio, fusion, routing, and I wonder what will happen if one of the components break down? It might be difficult to debug. Think about this. These and many more feedback below are designed to acknowledge these strengths and suggest areas for further refinement and depth.

i. Understanding of Problem Domain

The project is well-scoped and motivated, and identified a research area and demonstrated an understanding of the problem domain though this can be amplified more clearly. You conducted a critical analysis of literature within the context of the project to uncover current thinking when it comes to the related themes. However, you may need to conduct an exploratory analysis of existing studies that utilised similar approaches to produce findings that could inform your design choices for the project. Good job presenting how your project planning evolved in tabular form. It will also benefit

from a structured heuristic evaluation of existing related prototypes to enhance your own design requirements for the frontend. You would need to make more clearer how you identified how some of the legal, social, ethical and professional issues that are relevant to the project before discussing how they were addressed.

ii. Development of product and ideas

Some efforts here in terms of demonstration of complexity in the design and implementation of the system. While the project leveraged hybrid architecture featuring MobileNetV2, ResNet18, and ViT, along with a dynamic routing mechanism, this can be enhanced to also show how the data and model pipelines informed your system requirements and analysis. This will strengthen your discussion of the development process. This report will benefit from showing how several approaches were considered before settling for the one you used and justifications. Some thoughts showing how your project plans evolved would be useful. Good job describing how you resolved relevant legal, social, ethical and professional issues that might impact the system.

iii. Product build and evaluation

While the project shows good technical ability in building the product, it is not clear if and how hypotheses was tested, although you refined your research questions. While the report shows fewer discussions of model testing, however, you may have to enhance the system testing procedure (with test cases and corresponding findings) along with evaluation procedures with operational stakeholders (not just metrics) to show usability and appropriateness of the product for the identified problem area. It is also required that you clearly map the product in relation to original objectives and criteria. Also, it would benefit from a clearer and more detailed description of the product build cycle.

iv. Conclusions and critical review

The report demonstrates critical thinking in developing the product and writing up the project. While the limitation of the work is not really clear, it would be nice to enhance your discussion of the lessons learnt while completing the research, problems encountered and discussed how they were tackled, and mistakes made and how they were resolved. Also, it would benefit from a clearer reflection on how the student adjusted the project plans during the research. Although the report presents some recommendations and future work which are not elaborate, it would need to go beyond to show a clear vision – how the recommendations could be implemented in the real-world. Some enhancements in report formats and reference list may be required.

Comments on the viva/demonstration

Have you held a demonstration or viva for this project? YES/NO

Excellent presentation! The student took on his questions very well, demonstrating a good knowledge and skills in building his product and how it applies to real-world contexts.

Comments on the Standard of Presentation

Has the report met presentation criteria? YES/NO

The report met the presentation criteria.

Authenticity of the Submission

All final reports are uploaded into TurnItIn for plagiarism checking.

TurnItIn Percentage Match = %

Comments on the TurnItIn report, and any other plagiarism issues.

Marking Criteria

Each of the following four criteria carries equal weight and lists indicative points to consider. These may vary depending on the project's aims and objectives.
The marks entered average to the overall grade awarded.

1. Understanding of Problem Domain

This section is about assessing the ability to identify and investigate a suitable problem and follow an appropriate project methodology to solve the problem:

- Identified an area to research or investigate and a problem to be solved
- Demonstrated understanding of the problem domain
- Shown how project objectives were formed and project planning took place
- Discussed the research or investigation within the context of the project
- Critical evaluation (of the appropriateness) of the current thinking on the research area
- Compared similar products and systems
- Selected suitable criteria for development of product and ideas
- Chosen appropriate tools for modelling and development
- Production of requirements specification/client brief of suitable complexity
- Identified any legal, social, ethical and professional issues that are relevant to the project

	>89%	80-89%	70-79%	60-69%	50-59%	40-49%	<40
Report							
Demonstration							

Comments:

Report: Please see above for overall assessment of the project report.

Demonstration: Viva was conducted for this student's research project, and extensive feedback comments are provided for future enhancement.

2. Development of Product and Ideas

This section is about assessing the level of the development and the student's competency within the context of the chosen topic area:

- Use of suitable data collection techniques
- Use of suitable analysis tools to evaluate data
- Demonstrated complexity in the design and implementation of the product
- Discussed the development process
- Shown that a number of alternative approaches have been considered
- Explained the reasons for selecting a particular solution
- Demonstrated changes in the project plan (incremental development verified during supervision/ progress meetings and shared repositories)
- Resolution of relevant legal, social, ethical or professional issues

	>89%	80-89%	70-79%	60-69%	50-59%	40-49%	<40
Report							
Demonstration							

Comments:

Report: Please see above for overall assessment of the project report.

Demonstration: Viva was conducted for this student's research project, and extensive feedback comments are provided for future enhancement.

3. Product Build and Evaluation

This section is about assessing the ability to present the completed work and discuss issues of quality, usability, etc.:

- Demonstrated technical ability in building the product
- Demonstrated the full scope of product developed
- Clear plan of how the hypothesis might be tested
- In-depth investigation of the context/literature/other relevant products
- A reasoned thread going throughout the project
- Shown that the product has been tested and evaluated appropriately
- Discussed the quality of the product in relation to original objectives and criteria
- Demonstrated the usability and appropriateness of the product for the problem domain
- Identified where and how improvement can be made

	>89%	80-89%	70-79%	60-69%	50-59%	40-49%	<40
Report							
Demonstration							

Comments:

Report: Please see above for overall assessment of the project report.

Demonstration: Viva was conducted for this student's research project, and extensive feedback comments are provided for future enhancement.

4. Conclusions and Critical Review

This section is about assessing the student's ability to be critical of their own work and show reflective thinking:

- Demonstrated critical thinking in writing up the project
- Discussed lessons learnt whilst completing the project
- Identified any problems encountered and discussed how they were tackled
- Identified mistakes made and lessons learnt
- Reflected on how the project plan changed during the development
- Made suggestions as to how the work can be improved
- Identified how the project might be taken further or expanded

	>89%	80-89%	70-79%	60-69%	50-59%	40-49%	<40
Report							
Demonstration							

Comments:

Report: Please see above for overall assessment of the project report.

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